PHYS4055 Week 1 Tutorial Notes

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(Dated: July 28, 2026)

Question 1: Displacement of the annihilation operator

The displacement operator is defined by

$$\hat{D}(\alpha) = \exp\left(\alpha\hat{a}^\dagger - \alpha^*\hat{a}\right), \tag{1}$$

where α is a complex number and

$$[\hat{a}, \hat{a}^\dagger] = 1. \tag{2}$$

Use the Baker–Campbell–Hausdorff relation given by

$$\begin{aligned} e^{\hat{X}}\hat{Y}e^{-\hat{X}} = \hat{Y} &+ [\hat{X}, \hat{Y}] \\ &+ \frac{1}{2!}[\hat{X}, [\hat{X}, \hat{Y}]] \\ &+ \frac{1}{3!}[\hat{X}, [\hat{X}, [\hat{X}, \hat{Y}]]] + \cdots . \end{aligned} \tag{3}$$

for any two operators $\hat{X}$ and $\hat{Y}$, to show that

$$\boxed{\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha}. \tag{4}$$

Worked solution

Define

$$\hat{A} = \alpha\hat{a}^\dagger - \alpha^*\hat{a}. \tag{5}$$

The displacement operator can then be written as

$$\hat{D}(\alpha) = e^{\hat{A}}. \tag{6}$$

We calculate the Hermitian conjugate of $\hat{A}$:

$$\begin{aligned} \hat{A}^\dagger &= \left(\alpha\hat{a}^\dagger - \alpha^*\hat{a}\right)^\dagger && (7)\\ &= \alpha^*\hat{a} - \alpha\hat{a}^\dagger && (8)\\ &= -\hat{A}. && (9) \end{aligned}$$

Therefore,

$$\begin{aligned} \hat{D}^\dagger(\alpha) &= \left(e^{\hat{A}}\right)^\dagger && (10)\\ &= e^{\hat{A}^\dagger} && (11)\\ &= e^{-\hat{A}}. && (12) \end{aligned}$$

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The operator that we wish to calculate is therefore

$$\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = e^{-\hat{A}}\hat{a}e^{\hat{A}}. \tag{13}$$

To match the Baker–Campbell–Hausdorff relation, we identify

$$\hat{X} = -\hat{A} \tag{14}$$

and

$$\hat{Y} = \hat{a}. \tag{15}$$

The Baker–Campbell–Hausdorff relation then gives

$$\begin{aligned} e^{-\hat{A}}\hat{a}e^{\hat{A}} = \hat{a} &+ [-\hat{A}, \hat{a}] \\ &+ \frac{1}{2!}[-\hat{A}, [-\hat{A}, \hat{a}]] \\ &+ \cdots . \end{aligned} \tag{16}$$

We first calculate the first commutator:

$$[-\hat{A}, \hat{a}] = -[\hat{A}, \hat{a}]. \tag{17}$$

Using the definition of $\hat{A}$,

$$[\hat{A}, \hat{a}] = \left[\alpha\hat{a}^\dagger - \alpha^*\hat{a}, \hat{a}\right] \tag{18}$$

$$= \alpha[\hat{a}^\dagger, \hat{a}] - \alpha^*[\hat{a}, \hat{a}]. \tag{19}$$

The canonical commutation relation gives

$$[\hat{a}^\dagger, \hat{a}] = -1, \tag{20}$$

and

$$[\hat{a}, \hat{a}] = 0. \tag{21}$$

Therefore,

$$[\hat{A}, \hat{a}] = \alpha(-1) - \alpha^*(0) \tag{22}$$

$$= -\alpha. \tag{23}$$

It follows that

$$[-\hat{A}, \hat{a}] = -(-\alpha) \tag{24}$$

$$= \alpha. \tag{25}$$

We now calculate the second nested commutator:

$$[-\hat{A}, [-\hat{A}, \hat{a}]] = [-\hat{A}, \alpha]. \tag{26}$$

Since α is a complex number rather than an operator,

$$[-\hat{A}, \alpha] = 0. \tag{27}$$

All higher nested commutators also vanish. Therefore, the Baker–Campbell–Hausdorff expansion terminates after the first commutator:

$$e^{-\hat{A}}\hat{a}e^{\hat{A}} = \hat{a} + [-\hat{A}, \hat{a}] \tag{28}$$

$$= \hat{a} + \alpha. \tag{29}$$

Hence,

$$\boxed{\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha}. \tag{30}$$

Question 2: Coherent states as eigenstates

A coherent state is defined by

$$|\alpha\rangle = \hat{D}(\alpha)|0\rangle, \tag{31}$$

where $|0\rangle$ is the vacuum state.

Using the displacement-operator identity

$$\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha, \tag{32}$$

show that $|\alpha\rangle$ is an eigenstate of the annihilation operator $\hat{a}$.

Worked solution

We begin with the action of the annihilation operator on the coherent state:

$$\hat{a}|\alpha\rangle = \hat{a}\hat{D}(\alpha)|0\rangle. \tag{33}$$

We insert the identity operator in the form

$$\hat{I} = \hat{D}(\alpha)\hat{D}^\dagger(\alpha). \tag{34}$$

Therefore,

$$\hat{a}|\alpha\rangle = \hat{D}(\alpha)\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha)|0\rangle. \tag{35}$$

We now use

$$\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha. \tag{36}$$

This gives

$$\hat{a}|\alpha\rangle = \hat{D}(\alpha)\left(\hat{a} + \alpha\right)|0\rangle. \tag{37}$$

Expanding the right-hand side gives

$$\hat{a}|\alpha\rangle = \hat{D}(\alpha)\hat{a}|0\rangle + \alpha\hat{D}(\alpha)|0\rangle. \tag{38}$$

The annihilation operator acting on the vacuum gives

$$\hat{a}|0\rangle = 0. \tag{39}$$

Therefore,

$$\hat{D}(\alpha)\hat{a}|0\rangle = 0. \tag{40}$$

The remaining term is

$$\hat{a}|\alpha\rangle = \alpha\hat{D}(\alpha)|0\rangle. \tag{41}$$

Using

$$\hat{D}(\alpha)|0\rangle = |\alpha\rangle, \tag{42}$$

we obtain

$$\boxed{\hat{a}|\alpha\rangle = \alpha|\alpha\rangle}. \tag{43}$$

Therefore, the coherent state $|\alpha\rangle$ is an eigenstate of the annihilation operator $\hat{a}$, with eigenvalue α.

Question 3: Electric field of a coherent state

Consider a single travelling optical mode with electric-field operator

$$\hat{\mathbf{E}}(z,t) = \mathbf{e}E_0\left(\hat{a}e^{i\phi} + \hat{a}^\dagger e^{-i\phi}\right), \tag{44}$$

where

$$E_0 = \sqrt{\frac{\hbar\omega}{2\varepsilon_0 V}} \tag{45}$$

and

$$\phi = kz - \omega t. \tag{46}$$

Here, $\mathbf{e}$ is the polarization vector, V is the quantization volume, and ω is the angular frequency of the mode. For a coherent state satisfying

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle, \tag{47}$$

calculate the expectation value

$$\langle\hat{\mathbf{E}}(z,t)\rangle_\alpha = \langle\alpha|\hat{\mathbf{E}}(z,t)|\alpha\rangle. \tag{48}$$

Write

$$\alpha = |\alpha|e^{i\theta} \tag{49}$$

and show that the expectation value has the form of a classical oscillating electric field.

Worked solution

The coherent-state eigenvalue equation is

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle. \tag{50}$$

Taking the Hermitian conjugate of both sides gives

$$\left(\hat{a}|\alpha\rangle\right)^\dagger = \left(\alpha|\alpha\rangle\right)^\dagger. \tag{51}$$

Therefore,

$$\langle\alpha|\hat{a}^\dagger = \alpha^*\langle\alpha|. \tag{52}$$

We now calculate the expectation value of the electric field:

$$\langle\hat{\mathbf{E}}(z,t)\rangle_\alpha = \langle\alpha|\hat{\mathbf{E}}(z,t)|\alpha\rangle. \tag{53}$$

Substituting the electric-field operator gives

$$\langle\hat{\mathbf{E}}(z,t)\rangle_\alpha = \mathbf{e}E_0\langle\alpha|\left(\hat{a}e^{i\phi} + \hat{a}^\dagger e^{-i\phi}\right)|\alpha\rangle. \tag{54}$$

We separate the two terms:

$$\begin{aligned}\langle\hat{\mathbf{E}}(z,t)\rangle_\alpha = {}& \mathbf{e}E_0 e^{i\phi}\langle\alpha|\hat{a}|\alpha\rangle \\ & + \mathbf{e}E_0 e^{-i\phi}\langle\alpha|\hat{a}^\dagger|\alpha\rangle.\end{aligned} \tag{55}$$

For the first expectation value,

$$\langle\alpha|\hat{a}|\alpha\rangle = \langle\alpha|\left(\hat{a}|\alpha\rangle\right). \tag{56}$$

Using the coherent-state eigenvalue equation gives

$$\langle\alpha|\hat{a}|\alpha\rangle = \alpha\langle\alpha|\alpha\rangle. \tag{57}$$

Since the coherent state is normalized,

$$\langle\alpha|\alpha\rangle = 1. \tag{58}$$

Therefore,

$$\langle\alpha|\hat{a}|\alpha\rangle = \alpha. \tag{59}$$

Similarly,

$$\langle\alpha|\hat{a}^\dagger|\alpha\rangle = \alpha^*. \tag{60}$$

The electric-field expectation value is therefore

$$\langle\hat{\mathbf{E}}(z,t)\rangle_\alpha = \mathbf{e}E_0\left(\alpha e^{i\phi} + \alpha^* e^{-i\phi}\right). \tag{61}$$

We now write the coherent amplitude as

$$\alpha = |\alpha|e^{i\theta}. \tag{62}$$

Its complex conjugate is

$$\alpha^* = |\alpha|e^{-i\theta}. \tag{63}$$

The first term becomes

$$\alpha e^{i\phi} = |\alpha|e^{i(\phi+\theta)}. \tag{64}$$

The second term becomes

$$\alpha^* e^{-i\phi} = |\alpha|e^{-i(\phi+\theta)}. \tag{65}$$

Therefore,

$$\langle\hat{\mathbf{E}}(z,t)\rangle_\alpha = \mathbf{e}E_0|\alpha|\left(e^{i(\phi+\theta)} + e^{-i(\phi+\theta)}\right). \tag{66}$$

Using

$$e^{ix} + e^{-ix} = 2\cos(x), \tag{67}$$

we obtain

$$\langle\hat{\mathbf{E}}(z,t)\rangle_\alpha = 2\mathbf{e}E_0|\alpha|\cos(\phi+\theta). \tag{68}$$

Finally, substituting

$$\phi = kz - \omega t \tag{69}$$

gives

$$\boxed{\langle\hat{\mathbf{E}}(z,t)\rangle_\alpha = 2\mathbf{e}E_0|\alpha|\cos\left(kz - \omega t + \theta\right)}. \tag{70}$$

This expectation value has the same sinusoidal form as a classical travelling electromagnetic wave. The magnitude $|\alpha|$ determines the field amplitude, while the argument θ determines its phase.